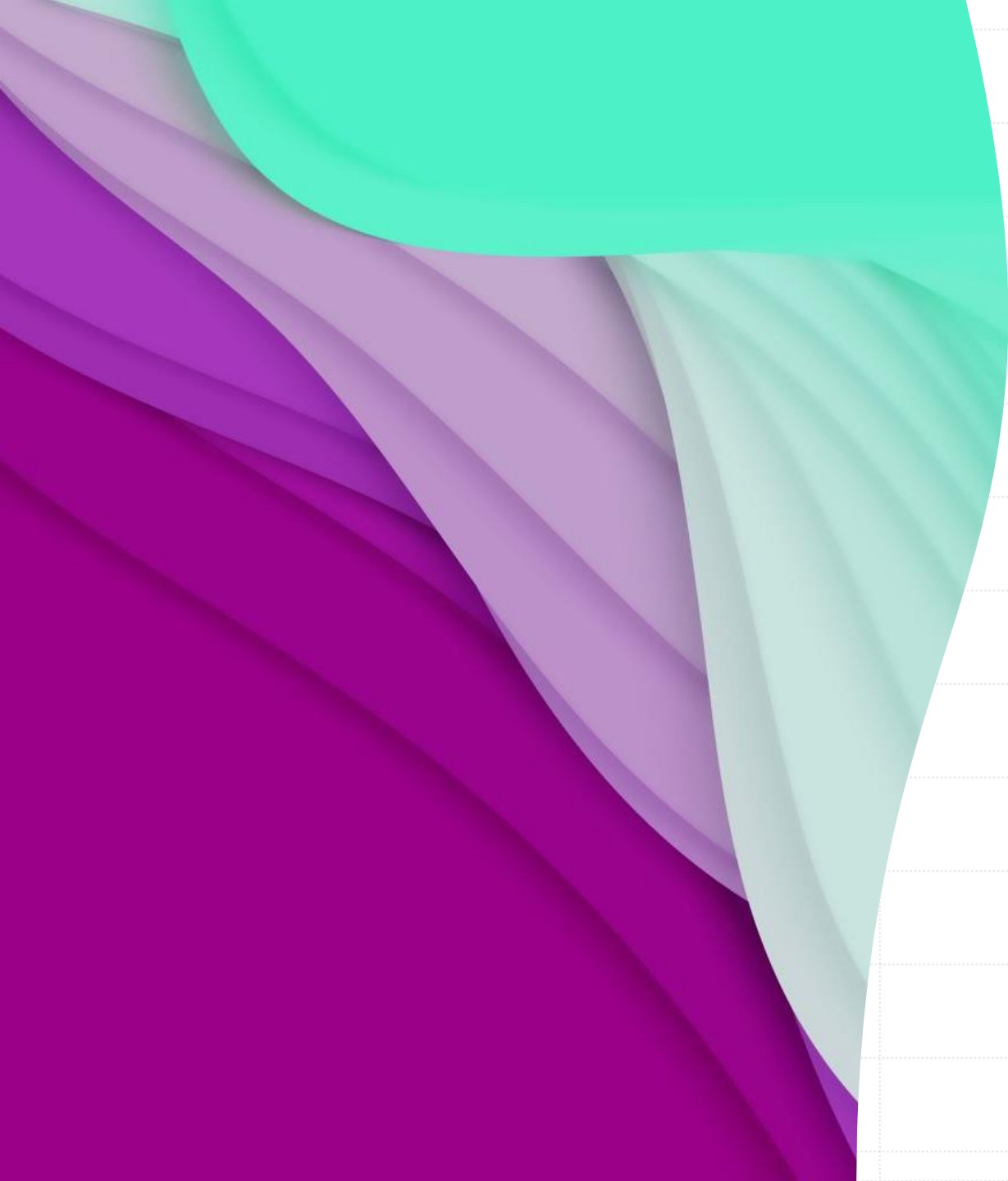




Python Programming Bootcamp – Week 3, Day 3

Introduction to Tkinter(GUI Applications) in
Python

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What is Tkinter?

- **Definition:** **Tkinter** is the standard GUI (Graphical User Interface) library for Python.
- **Uses of Tkinter:** Building **desktop applications** with graphical elements like windows, **buttons**, **labels**, etc.
- **Advantages of Tkinter:**
 - Simple and easy to use.
 - Cross-platform.
 - Included in Python's standard library.



Tkinter Installation and Setup

- Installing Python (if not installed).
- Tkinter comes pre-installed with Python, no need for separate installation.
- Checking Tkinter installation: `import tkinter`



Basic Tkinter Window

- **Overview** of creating the first window in Tkinter.

```
import tkinter as tk

# Create a root window
root = tk.Tk()

# Window title
root.title("First Tkinter Window")

# Run the window
root.mainloop()
```



Widgets in Tkinter

- Briefly introduce the most common widgets:
 - Label, Button, Entry, Text, Canvas, Frame, Checkbutton, Radiobutton.



Hands-on Activity 1

- Create a basic Tkinter window with a label and a button

```
import tkinter as tk
def greet():
    print("Hello, welcome to Tkinter!")
root = tk.Tk()
# Label Widget
label = tk.Label(root, text="Welcome to Tkinter")
label.pack()
# Button Widget
button = tk.Button(root, text="Greet", command=greet)
button.pack()
root.mainloop()
```





Working with Tkinter Widgets



Tkinter Widgets in Depth

- Discuss how to use and configure widgets in [Tkinter](#).
 - **Label** for displaying text.
 - **Button** for triggering actions.
 - **Entry** for text input.
 - **Text** for multi-line text input.

Adding Labels and Entry Widgets

```
import tkinter as tk
root = tk.Tk()
tk.Label(root, text="Username").grid(row=0)
tk.Label(root, text="Password").grid(row=1)
tk.Button(root, text = "Submit").grid(row=3)
# Entry Widgets
username = tk.Entry(root)
password = tk.Entry(root, show="*")
# Bind User name to Grid
username.grid(row=0, column=1)
password.grid(row=1, column=1)

root.mainloop()
```

Layout Management in Tkinter



Introduction to Tkinter Layout Managers

- **pack()**: Simple layout method.
- **grid()**: More control over positioning.
- **place()**: Absolute positioning

Hands-on Activity 3

- Create a form with multiple widgets using the grid() layout method.



Advanced Tkinter Widgets



Working with Frames

Frame is used to group widgets together and manage complex layouts.

```
import tkinter as tk
root = tk.Tk()
frame = tk.Frame(root)
frame.pack()

tk.Label(frame, text="Inside Frame").pack()
```

Using Listbox and Scrollbars

Example of how to create a [Listbox](#) and add a [scrollbar](#) to it.

```
import tkinter as tk
root = tk.Tk()
listbox = tk.Listbox(root)
listbox.pack()

for item in ["Apple", "Banana", "Cherry", 1, 2, 2, 3, 4, 5, 6, 7, 7, 7, 8, 9, 7, 65, 55]:
    listbox.insert(tk.END, item)

scrollbar = tk.Scrollbar(root)
scrollbar.pack(side=tk.RIGHT, fill=tk.Y)
root.mainloop()
```

Canvas and Drawing Shapes

Introduction to using [Canvas](#) for drawing shapes like rectangles, circles, and lines.

```
import tkinter as tk
root = tk.Tk()
canvas = tk.Canvas(root, width=200, height=100)
canvas.pack()

canvas.create_rectangle(50, 20, 150, 80, fill="blue")

root.mainloop()
```


Hands-on Activity 4

- Create a simple canvas and draw shapes like circles, rectangles, and lines.





Questions?